

# Building Legal RAG: what each component contributes

## 01 CONTEXT AND HYPOTHESIS

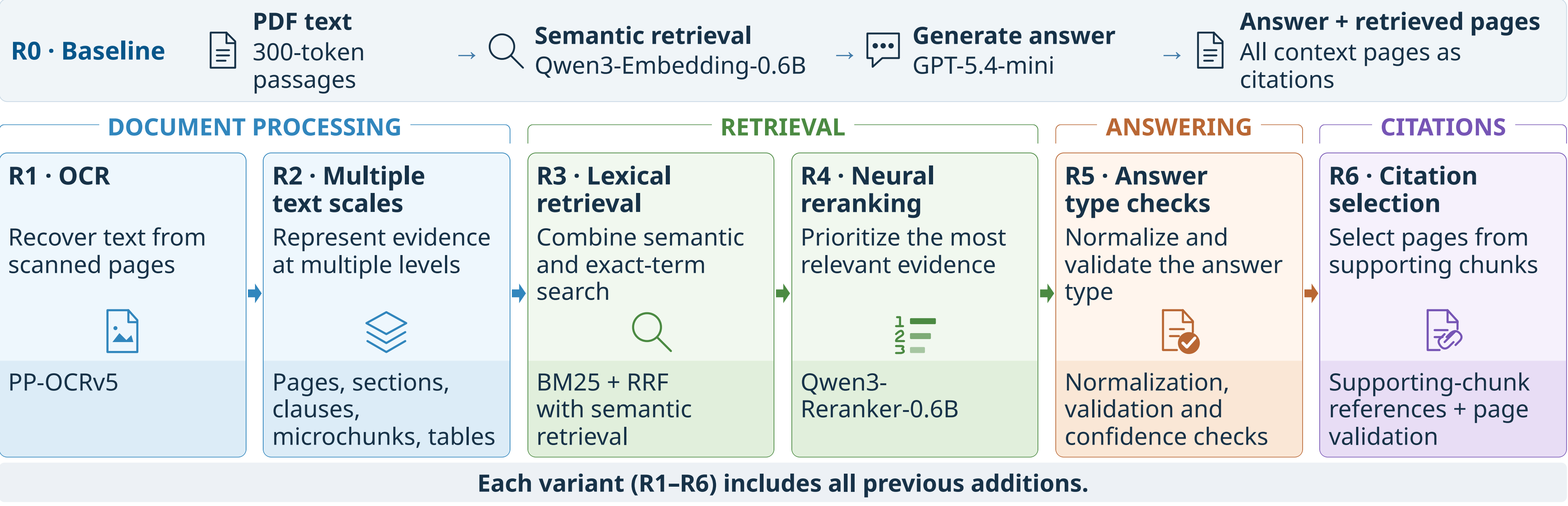
Retrieval-augmented generation (RAG) grounds language-model answers in external documents [1]. Legal QA needs correct answers and precise, complete source-page citations [2, 3]. PDFs may lack searchable text; evidence spans different text scales and may require exact legal terms. Retrieving more passages can also introduce irrelevant pages.

How can a legal RAG pipeline improve answer quality while keeping source-page citations both precise and complete?

|                                                                                                                                 |                                                                                                                                          |                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>H1 · Retrieval</b><br>Multiple text scales, hybrid search and reranking will improve answers and evidence retrieval over R0. | <b>H2 · Scan robustness</b><br>Recovering text from scanned documents will reduce the performance loss caused by the missing text layer. | <b>H3 · Citation selection</b><br>Selecting pages after answer generation will improve citation precision while retaining most of the relevant evidence. |
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## 02 METHODOLOGY

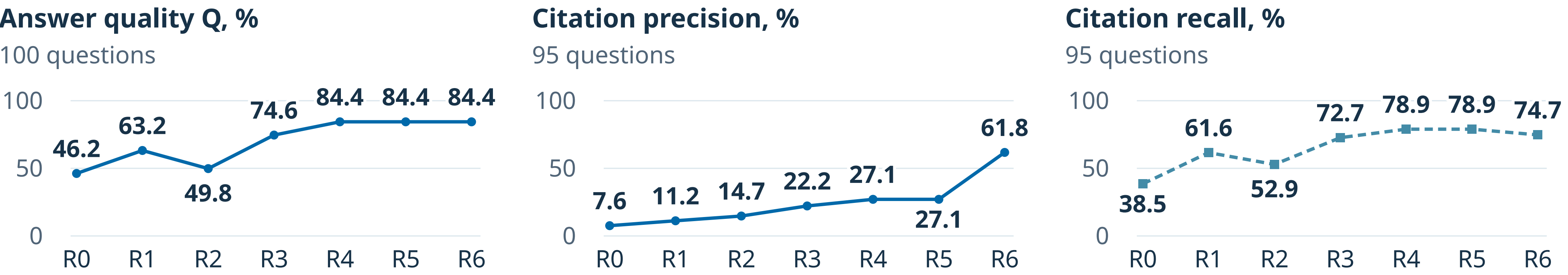
**Data:** 30 DIFC legal PDFs (590 pages), 100 questions: 70 structured-answer and 30 free-text. Reference source pages are available for 95 questions. In the mixed corpus, 10 PDFs (107 pages) become synthetic scans: page images at 300 DPI with the text layer removed.



**Answer generation:** A fixed GPT-5.4-mini configuration is used across variants, so changes in performance reflect modifications to the surrounding pipeline rather than a change of the generating LLM.

|                                                                                                                               |                                                                                                                             |                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Answer quality</b><br><i>Is the generated answer correct?</i><br><b>Q = 0.7 × structured score + 0.3 × free-text score</b> | <b>Citation precision</b><br><i>Are the cited pages actually relevant?</i><br>Share of cited pages that support the answer. | <b>Citation recall</b><br><i>Did the system cite all required evidence?</i><br>Share of annotated supporting pages that were cited. |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

## 03 RESULTS



|                                                      |      |
|------------------------------------------------------|------|
| <b>OCR recovers performance on scanned documents</b> |      |
| Original corpus · R0                                 | 65.4 |
| Mixed corpus (original + synthetic scans) · R0       | 46.2 |
| Mixed corpus + OCR · R1                              | 63.2 |

Converting 10 documents into synthetic scans reduces answer quality from 65.4% to 46.2%. OCR restores most of this loss, reaching 63.2%.

|                                                                                                                          |
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| <b>Hypothesis evaluation</b>                                                                                             |
| <b>H1 · Partially supported.</b> Multiple scales alone reduce Q; lexical retrieval and reranking improve Q and recall.   |
| <b>H2 · Supported.</b> OCR restores most of the loss caused by removing the text layer.                                  |
| <b>H3 · Supported with a trade-off.</b> Precision rises from 27.1% to 61.8%; recall falls from 78.9% to 74.7% (–4.2 pp). |

**Limitations**  
one DIFC corpus and 100 questions; one run per condition with no significance testing; LLM-based evaluation of free-text answers.

**Conclusion**  
Legal RAG should be designed as a multi-stage pipeline: OCR improves robustness to unreadable PDFs, hybrid retrieval and reranking improve answer quality, and post-answer citation selection improves citation precision. However, citation selection introduces a precision-recall trade-off, so answer quality and citation quality should be optimized and evaluated separately.